**SIMATS ENGINEERING**

**CO1 - ASSESSMENT TOOL 2**

**ERROR ANALYSIS TASK**

| Name of the Student | Narra Sai Mounika |
|---|---|
| Reg. No | 192425367 |

**COURSE NAME:** **Deep Learning** **COURSE CODE:** **MLA0403**
**FACULTY : Dr.Arul Raj A.M**

---

**COURSE OUTCOME COVERED**

**CO 1:** Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BTL-4)

**CO1 Weightage: 15%**

**Assessment Tool Weightage: 30%**

**Duration: 30 minutes**

**Marks: 25**

---

**Code of Conduct:**

I Narra Sai Mounika Reg no: 192425367 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

N. Sai Mounika
**Signature of the student:**

**QUESTIONS:** **Total Marks:25**

## 1. Error Analysis in Maximum Likelihood Estimation (MLE)

A student builds a binary classifier using Maximum Likelihood Estimation. The likelihood function is incorrectly written as the sum of probabilities instead of the product. The model produces poor parameter estimates even with large data.

**Tasks:**

a) Identify the conceptual error in the likelihood formulation.

**Answer :** The likelihood function was incorrectly written as the sum of probabilities. In MLE, the likelihood should be the product of probabilities of all independent observations.

b) Explain how this mistake affects parameter estimation.

**Answer :** Using the sum does not represent the joint probability of the data. As a result, the optimization process estimates incorrect parameters, leading to poor model performance even with large datasets.

c) Suggest the correct mathematical formulation and reasoning.
**Answer :** For independent observations:
L(θ)=∏i=1nP(xi∣θ)L(\theta)=\prod_{i=1}^{n} P(x_i|\theta)L(θ)=i=1∏nP(xi∣θ)The product is used because the probability of observing all samples together equals the multiplication of individual probabilities.

d) How would using log-likelihood fix computational issues?
**Answer :** Log-likelihood converts multiplication into addition:
logL(θ)=∑i=1nlogP(xi|θ)\log L(\theta)=\sum_{i=1}^{n}\log P(x_i|\theta)logL(θ)=i=1∑nlogP(xi∣θ)
**Benefits:**

- Prevents numerical underflow.
- Simplifies differentiation.
- Improves optimization efficiency.

## 2. Error Diagnosis in Perceptron Learning

While implementing a Perceptron, a student observes that the model fails to converge even after many iterations on a linearly separable dataset.
**Tasks:**
a) List possible implementation errors in the learning algorithm.
**Answer :**

- Incorrect weight update formula.
- Missing bias term.
- Wrong activation function.
- Incorrect classification threshold.
- Improper data preprocessing.
- Coding mistakes in update loops.

b) Analyze how incorrect learning rate or weight updates affect convergence.
**Answer :**

- A very high learning rate causes oscillations.
- Very low learning rate slows learning.
- Incorrect weight updates move the decision boundary in the wrong direction, preventing convergence.

c) Identify dataset-related issues that may cause this problem.
**Answer :**

- Data may not be truly linearly separable.
- Presence of noisy samples.
- Incorrect labels.
- Unscaled feature values.

d) Propose modifications to ensure convergence.
**Answer :**

- Normalize features.
- Verify correct labels.
- Use the proper learning rate.
- Include bias term.
- Apply the standard Perceptron update rule:

w=w+η(y−y^)xw = w + \eta(y-\hat{y})xw=w+η(y−y^)x

## 3. Backpropagation Gradient Error Analysis

A Multilayer Perceptron trained using Backpropagation shows increasing loss during training instead of decreasing.

**Tasks:**

a) Identify possible mathematical or coding errors in gradient computation.

**Answer :**

- Incorrect derivative calculations.
- Wrong chain rule implementation.
- Sign errors in gradient updates.
- Updating weights before computing all gradients.
- Incorrect loss function implementation.

b) Explain the role of activation functions in gradient flow.

**Answer :**

Activation functions determine how gradients pass through layers.

- Sigmoid and Tanh may cause vanishing gradients.
- ReLU allows stronger gradient flow and faster learning.

c) Analyze how vanishing or exploding gradients affect training.

**Answer :**

**Vanishing Gradients:** Updates become extremely small, slowing learning.
**Exploding Gradients:** Updates become excessively large, causing instability and increasing loss.

d) Suggest techniques to fix the issue (e.g., normalization, initialization).

**Answer :**

- Use ReLU or Leaky ReLU.
- Apply Batch Normalization.
- Use Xavier or He Initialization.
- Gradient Clipping.
- Proper learning rate selection.

## 4. Gradient Descent vs SGD Error Investigation

A model trained using Gradient Descent converges very slowly, while the same model with Stochastic Gradient Descent shows unstable loss fluctuations.

**Tasks:**

a) Explain why batch gradient descent is slow in large datasets.

**Answer :** Batch Gradient Descent computes gradients using the entire dataset before each update. For large datasets, computation becomes expensive and training slows significantly.

b) Analyze the cause of fluctuations in SGD.

**Answer :** SGD updates parameters using one sample at a time. Different samples produce different gradients, causing noisy and fluctuating loss values.

c) Compare the error behavior of Mini-Batch Gradient Descent.

**Answer :**

Mini-Batch Gradient Descent uses a small group of samples.

- More stable than SGD.
- Faster than Batch Gradient Descent.
- Produces smoother convergence with moderate fluctuations.

d) Recommend the best approach and justify with error trade-offs.
**Answer :**
Mini-Batch Gradient Descent is generally preferred because:
- Faster than Batch GD.
- More stable than SGD.
- Efficient memory usage.
- Good balance between speed and convergence accuracy.

## 5. Curse of Dimensionality Error Analysis

A machine learning model built using high-dimensional data performs well on training data but poorly on test data due to overfitting, related to the Curse of Dimensionality.
**Tasks:**
a) Explain how high dimensionality increases model error.
**Answer :**
As dimensions increase:
- Data becomes sparse.
- More training samples are required.
- The model learns noise instead of useful patterns.
- Generalization error increases.

b) Identify signs of overfitting in this scenario.
**Answer :**
- Very high training accuracy.
- Low testing accuracy.
- Large gap between training and testing performance.
- Poor performance on unseen data.

c) Analyze why distance-based learning fails in high dimensions.
**Answer :**
In high-dimensional space:
- Distances between points become similar.
- Nearest neighbors become difficult to distinguish.
- Similarity measures lose effectiveness.
- Algorithms like KNN perform poorly.

d) Suggest dimensionality reduction or regularization techniques to improve performance.
**Answer :**

**Dimensionality Reduction:**

- Principal Component Analysis (PCA)
- Autoencoders
- Feature Selection

**Regularization Techniques:**

- L1 Regularization (Lasso)
- L2 Regularization (Ridge)
- Dropout
- Early Stopping

**RUBRICS FOR CALCULATION:**

| **Criteria** | **Weightage** | **Level 4 – Exemplary** | **Level 3 – Proficient** | **Level 2 – Developing** | **Level 1 – Beginning** |
|---|---|---|---|---|---|
| Understanding of Problem Context | 20 | Clearly understands the problem and accurately identifies all requirements | Good understanding with minor gaps | Basic understanding; some requirements unclear | Poor understanding or misinterpretation of the problem |
| Application of Concepts | 30 | Accurately applies appropriate concepts to solve complex problems | Applies relevant concepts correctly with minor errors | Applies basic concepts with limited accuracy | Fails to apply appropriate concepts |
| Problem-Solving Approach | 20 | Logical, well-structured, and efficient approach | Mostly logical approach with minor gaps | Basic approach with some inconsistencies | Illogical or unclear approach |
| Accuracy of Solution | 20 | Completely correct solution with proper justification | Mostly correct with minor errors | Partially correct solution | Incorrect or incomplete solution |
| Clarity | 10 | Clear, well-organized, and properly explained solution | Understandable with minor clarity issues | Limited clarity and explanation | Poorly presented and difficult to understand |